higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING (CAD COMPONENT)

(Second Paper)

NQF LEVEL 3

(12050663)

4 March 2019 (X-Paper)
09:00–11:00

REQUIREMENTS: **A4 drawing sheets**
Yellow cover folder (BOE 8/15)
Hot-rolled structural steel sections (BOE 8/2)

Calculators may be used.

This question paper consists of 4 pages and 2 diagram sheets.

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Rule off across the page on completion of each question.
6. Drawings and sketches must be clear enough to show detail.
7. Fully dimension and label ALL drawings.
8. Insert ALL loose DRAWING SHEETS in the folder. Do NOT staple.
9. Work neatly.

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QUESTION 1: REINFORCED-CONCRETE DETAILING

A 400 mm circular concrete column with EIGHT Y20 longitudinal bars and R6 helical binders at 150 mm centre must be constructed.

Draw, to scale 1:5, a vertical longitudinal section and a horizontal cross section through the column to show the constructional details. Show at least 500 mm of the length of the column and label the drawing. **[20]**

QUESTION 2: STRUCTURAL STEEL WORK DETAILING

2.1 Draw, to scale 1:1, a fully labelled section through a rolled steel angle 100 × 100 × 8. Neatly print the size and scale of the profile centred beneath the drawing. (10)

2.2 Refer to FIGURE 1, DIAGRAM SHEET 1 (attached), to complete the following question.

Do NOT copy the given view. Draw, to scale 1:5, the left view in the direction B in first-angle orthographic projection using CAD. Show any suitable simple design for reinforcement detail for the foundation pad.

Use the following information:

Top beam:	2Y12
Bottom beam:	2Y12
Stirrup rods:	12R10
Concrete beam:	50 mm
Diameter of holes for base plate:	24 mm

Insert ALL dimensions and related notes for the fabricator. (20)
[30]

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QUESTION 3: STRUCTURAL STEEL WORK DETAILING

3.1 Refer to FIGURE 2, DIAGRAM SHEET 2 (attached), and draw, to scale 1:5, typical bolted details at A.

Use the following specifications:

Top and bottom boom:	100 × 100 × 10 mm hot-rolled steel angle profiles
ALL internal members:	70 × 70 × 8 hot-rolled steel angle profiles
Gusset plates:	10 mm thick
Bolt hole centres:	100 mm
Edge distances:	35 mm

ALL bolt holes are drilled on the centre lines of the angles and ALL bolts are M20. Insert suitable dimensions. (15)

3.2 FIGURE 3, DIAGRAM SHEET 2 (attached), shows a line diagram (centre lines) of the front view of a typical steel roof truss to be used over a small workshop.

Draw, to scale 1:5, the front view of the joint at A. The FOUR members are connected by a 10 mm gusset plate.

Use the following specifications:

Roof pitch (slope):	As indicated on diagram sheet
Top chord members:	95 × 65 × 8 unequal-leg angles
Struts:	80 × 80 × 8 equal-leg angles
Gusset plate:	10 mm thick

Do NOT show any bolts or bolt holes. Clearly show the position of the gusset plate, centre lines and ALL labels. (10)
[25]

TOTAL: 75

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DIAGRAM SHEET 1

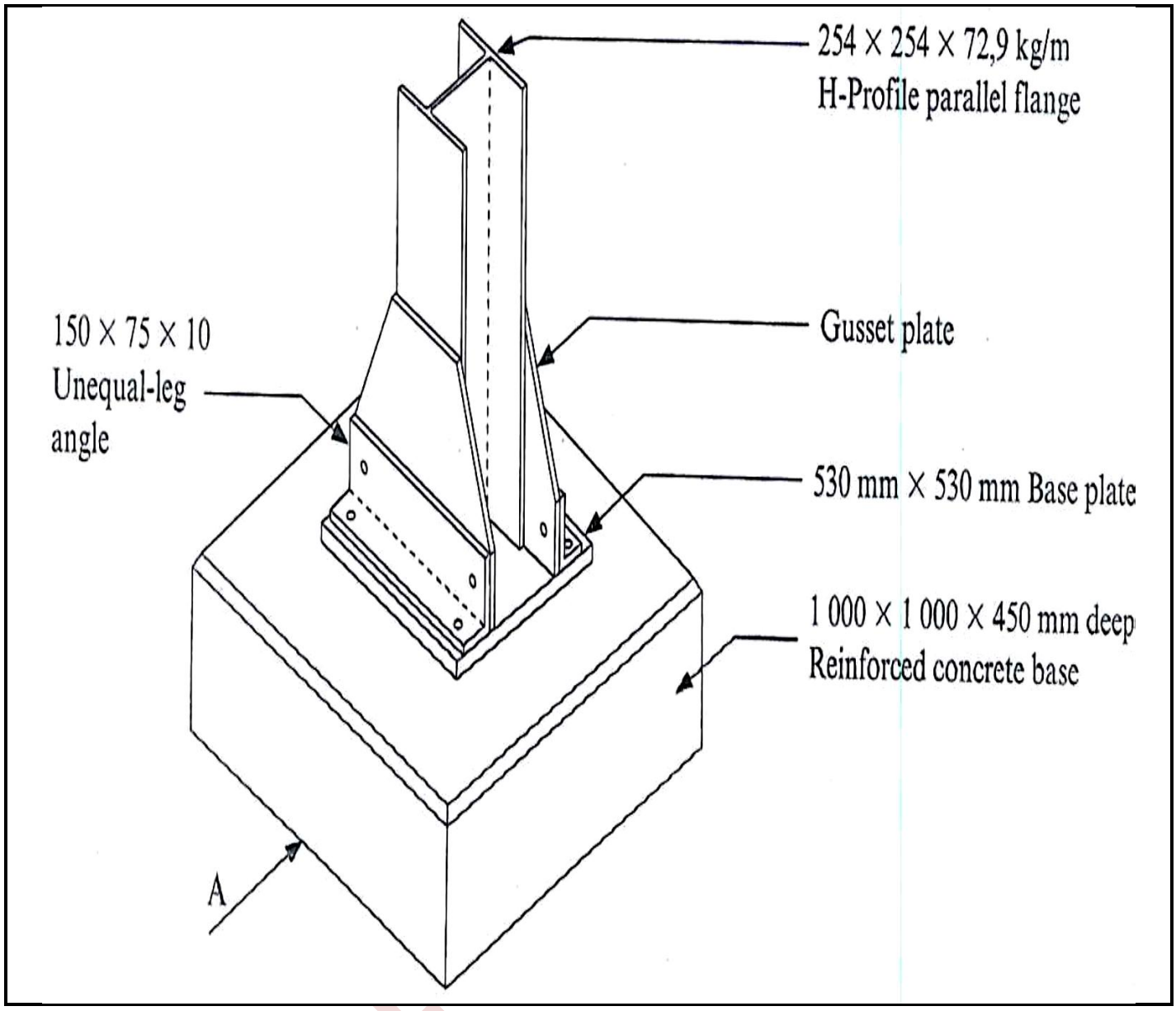


FIGURE 1

DIAGRAM SHEET 2

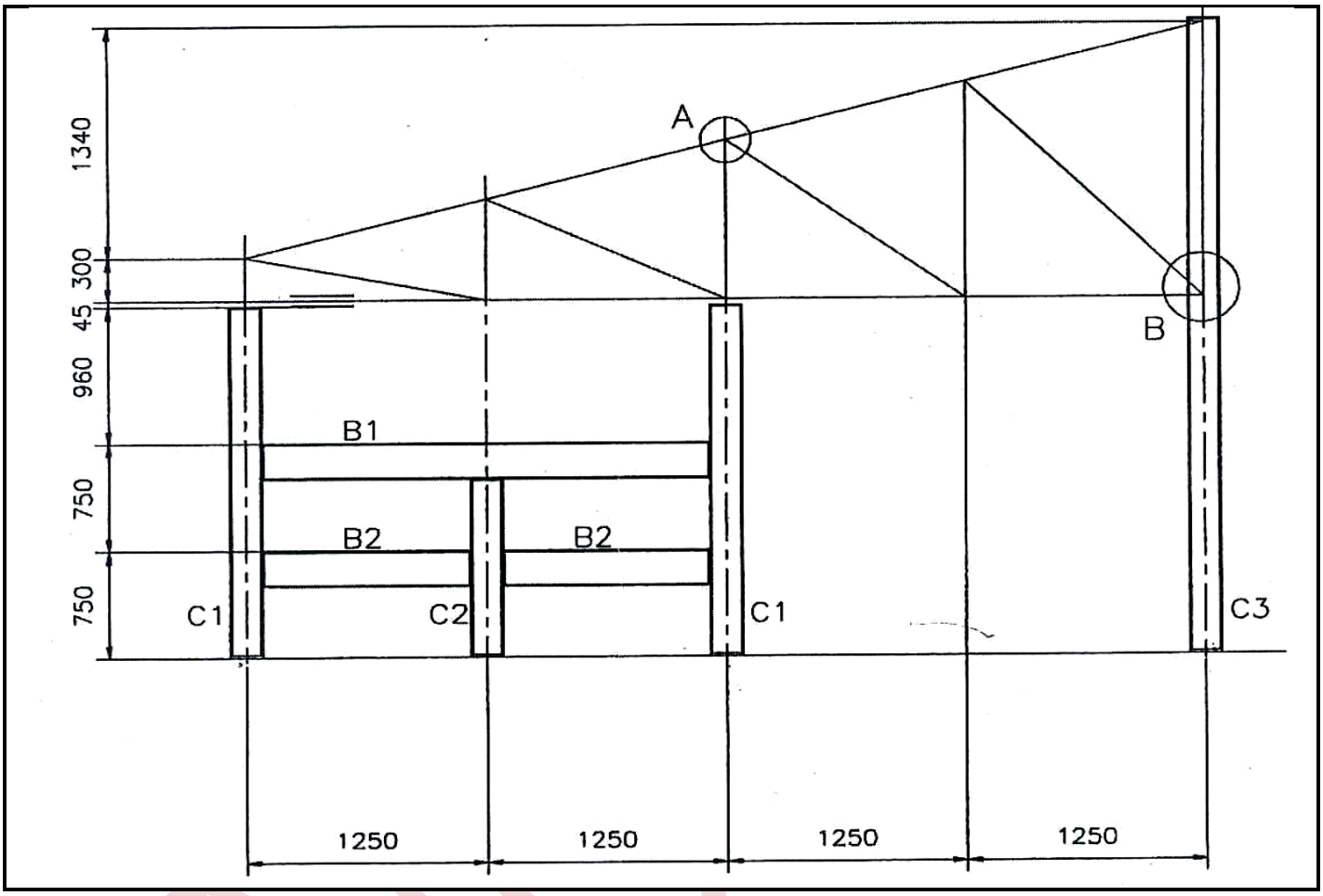


FIGURE 2

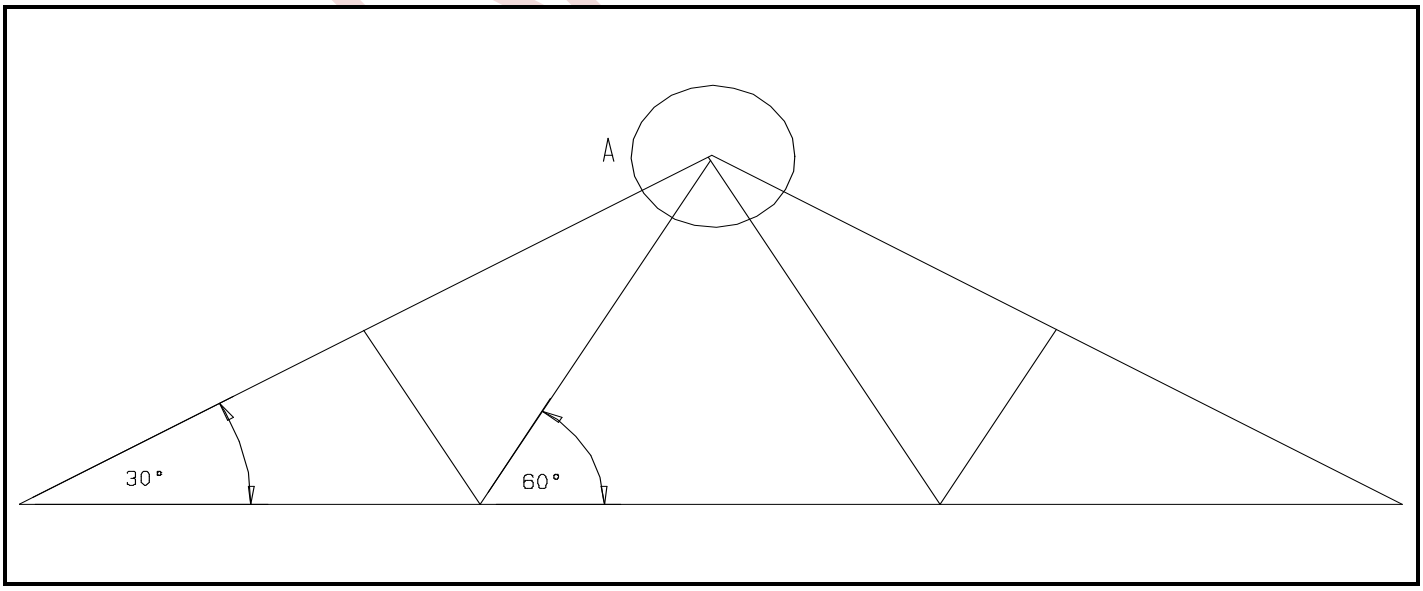


FIGURE 3

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